

Hearing in primitive mammals: *Monodelphis domestica* and *Marmosa elegans*

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Abstract

Although opossums of the Family Didelphidae usually serve as a parsimonious starting point for tracing the otological and neurological evolution of modern mammals, audiological data for Didelphid opossums is available only for the North American opossum (*Didelphis virginiana*) which because of its large size, may be one of the least representative genera of the family. The present report extends the audiological data to two other species of Didelphid opossums, *Monodelphis domestica*, and *Marmosa elegans*. At 60 dB SPL, the hearing of *Monodelphis* extends from 3.6 kHz to 77 kHz, with a range of best sensitivity from 8 to 64 kHz while the hearing of *Marmosa* extends from 3.8 kHz to 80 kHz, with a range of best sensitivity from 8 to 64 kHz. Neither species was found to be particularly sensitive to tones, with the average lowest threshold near 20 dB SPL for *Monodelphis* and 33 dB SPL for *Marmosa*. These results indicate that like the North American opossum both genera are sensitive to high frequencies yet relatively insensitive to sound. Because the hearing of the three genera of Didelphids agree in several respects, it can be concluded that sensitivity to high frequencies almost certainly was present in ancient mammals, probably following quickly after the acquisition of a 3 ossicle middle ear linkage. It is not unlikely that the utility value of high frequency hearing, rather than highly sensitive hearing, may have been a primary source of selective pressure for this morphological transformation.

Key words: *Monodelphis*; *Marmosa*; Didelphidae; Audiogram; Tone-detection threshold; Evolution

1. Introduction

Comparative otology and comparative neurology have shown that the structure of the marsupial opossum ear and brain closely approximates the form which can serve as a parsimonious starting point for tracing the evolution of modern viviparous mammals, whether marsupial or placental (Smith, 1910; Loo, 1930; Edinger, 1948; Simpson, 1949, 1959; Watson, 1953; Tumarkin, 1955; Olson, 1959; Fleischer, 1978; Kudo et al., 1986; Frost and Masterton, 1992). Fossil and geological evidence has corroborated this view: opossums virtually identical in bony structure to those extant today have survived at least since the Cretaceous and perhaps the Jurassic, 100 million or more years ago (Gregory, 1929). Apparently, the small opossum-like mammals first appeared in southern South America while that continent was still attached to Australia.

The opossums then dispersed throughout both of these pre-continentals. Later, the opossums also dispersed into Central and North America after they joined South America. Most of these marsupial radiations were later replaced by the evolving placentals leaving only opossums of the Family Didelphidae in the Americas. However, after the early dispersion of the opossums into precontinental Australia, the subsequent separation of Australia from South America protected the Australian radiation from placental competition resulting in the widely varied families of marsupials on that continent thereafter.

While the family of Didelphids is generally accepted as the best starting point among extant mammals for study of otological evolution, audiological data for Didelphid opossums is available only for the North American opossum, *Didelphis virginiana* (Ravizza et al., 1969), which with the largest body, head, brain and cochlea, is possibly the least representative of this family's members. For decades this lack of behavioral data was usually explained by reference to the difficul-

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ties of training recalcitrant animals. However, with the rapid development of behavioral technology over the last two decades, it has become possible to obtain reliable behavioral data on most if not all mammals, even those such as opossums, previously considered to be intractable.

The present report extends the audiological data to two other species of Didelphid opossums which, by being 'pouchless', are probably still more primitive than the North American opossum and also, more representative of the family (Kirsch, 1977; Marshall, 1979; Reig et al., 1987). *Monodelphis domestica* is a small (~ 90 grams) terrestrial marsupial inhabiting the neotropical region of South America. *Marmosa elegans* is a still smaller (~ 50 grams), partly arboreal, marsupial inhabiting the dry areas of the southern Andes – the area closest to the geographical origin of the entire sub-class of Marsupials.

2. Materials and methods

Five Brazilian opossums (*Monodelphis domestica*) and two Elegant opossums (*Marmosa elegans*) were trained and tested for their thresholds for pure tones. The behavioral and psychophysical methods have been described in detail elsewhere (e.g., Masterton and Granger, 1988, 1991). Briefly each opossum received its daily food supply only in the behavioral test apparatus. In the test apparatus, a mixture of commercial fox food blended into strained beef baby food was supplied at the rate of approximately 3 ml/min. This food slurry was delivered via a small spout through vinyl tubing driven by a syringe-pump outside of the sound chamber (Thompson et al., 1990). The food dispenser was operated by the opossum itself – contact with the food spout was sensed electronically and throughout the testing session, the animal's contact with the spout produced a constant stream of food without further contingency. Each of the daily sessions continued until the opossum was entirely satiated, making no further contact with the spout for five continuous minutes. This prolonged licking of food from the spout served to stabilize the opossum's head and ears in the sound field. Later, when the opossums had learned that the salient 'warning' sounds were always emitted by a speaker in a fixed direction, the pinna of both ears remained fixated on the speaker throughout the behavioral session. The hearing tests were imposed on this relatively constant behavioral background.

Details of threshold testing

A conditioned avoidance procedure was used in order to provide repeated threshold measurements without contamination from concurrent extinction of habituation of the response (e.g., see Masterton et al.,

1969; Heffner and Heffner, 1985b). In this procedure, each daily session was divided into 3-second periods. Each 3-second period was randomly predesignated as either a 'safe' or a 'warning' period with the warning periods distributed from 2 to 22 periods (i.e., 6 to 66 seconds) apart. During each safe period, a 'safe' signal was presented and during each warning period, a 'warning' signal was presented. For the hearing tests reported here, the safe signal was silence while the warning signal was a tone at one or another precalibrated intensity.

During the safe periods, there was no contingency on the opossum's licking or not licking from the spout. However, at the end of each warning period, a mild shock was administered from the floor to the spout. The shock was instantly escapable and for an audible sound, it was also avoidable merely by breaking contact with the food spout. The intensity of the shock was carefully titrated for each individual opossum to the minimum level that would result in its reliable avoidance when warned by a clearly audible tone (60 dB above normal threshold) and in no case did it exceed 1.0 mA. nor 100 microcoulombs. After a few days of warning sound-shock training, the opossums ceased licking and broke contact with the spout whenever the warning signal was presented, avoiding the shock entirely. For psychophysical testing, this breaking of contact with the spout was used as evidence that the warning signal had been detected.

Three arbitrary but unambiguous definitions of 'threshold' were routinely calculated (cf. Masterton and Granger, 1988, 1991). Although these differing definitions yielded slightly differing thresholds, each led to the same conclusion. Here, we refer only to the most rigorous one: The lowest intensity which was detected at a statistically reliable rate ($P < 0.05$). Once an animal appeared to have reached its threshold, testing for that frequency continued for 5 or more sessions.

Details of the behavioral apparatus

To minimize reflective surfaces and standing waves, the opossums were tested in a cage constructed of 2 inch $\times$ 4 inch hardware cloth placed in an acoustical chamber whose walls, ceiling, and floor were draped with loose burlap. The cage was mounted on foam rubber pads at a fixed distance from the sound speaker which was suspended from one corner of the chamber.

The food spout was fixed just outside the cage at its end near the speaker and a hole in the cage wall somewhat larger than the largest opossum's head allowed access to the spout. The cage, cage opening, and food spout were positioned in a manner that allowed easy access to the food only when the opossum's head (and hence, ears) were directed toward the speaker. By this means the sound-field in the immediate vicinity of

the opossum's head could be precisely controlled and accurately measured for each tone.

A microcomputer with appropriate interfacing was used to record the opossum's contact with the food spout, to control the occurrence of safe or warning signals, the intensity of the sounds and to determine 'hits' (correct detections), 'misses' (non-detections) and the rate of 'false-alarms' (false detections). Because the computer recorded the opossum's contact or non-contact with the food spout every tenth of a second during the last 1 second of every safe period as well as every warning period throughout the session, a refined and exact measure of the probability of a false-alarm (breaking of spout-contact during the safe signal) could be made. This probability, estimated over the previous 32 periods and up-dated after each safe period, was used (on-line) to determine if a given response to the warning signal was probably a 'hit' or a 'false-alarm'.

A low rate of false-alarms was maintained throughout the session by turning off the entire apparatus for 1 min whenever the rate of total cessations of licking during the most recent 32 safe periods rose above 0.1 (that is, more than 3 in 32 safe periods). After this 'time-out', the sequence of safe and warning signals was resumed as before though the responses to the warning signals were discounted until the false-alarm rate fell below 0.1 once more. With this contingency on false-alarm rate, a 'hit' was recorded whenever the probability of the opossum's response during the warning period relative to the same response during the previous 32 safe periods was less than 0.1. A 'miss' was recorded whenever the same probability was greater than 0.1.

Throughout each session the computer tracked or titrated the opossum's threshold by controlling the intensity of the sound for a given warning period on the basis of the behavioral outcome of the previous warning period. After a set of warm-up trials where the intensity was changed in 6 dB steps, the intensity was stepped down by 3 dB after each 'hit' and stepped up by 3 dB after each 'miss'. In this manner the computer stepped down to then around the opossum's threshold for the tone throughout a session in somewhat the same manner as in Bekesy audiometric testing of otological patients.

Details of sound generation, control and measurement

Tones were generated by a conventional sine-wave generator and their intensities controlled by the computer through a digital attenuator. The electrical signals began with the generator and were led in succession to the digital attenuator, an electronic switch which gated the signals with a rise-time of 50 ms, an amplifier, a fixed attenuator and finally, a wide-range coaxial speaker.

The entire sound system was calibrated and rou-

tinely recalibrated for each stimulus throughout the 6 month behavioral testing period. For calibration, a Bruel and Kjaer 1/4 inch microphone, preamplifier, octave filter set, and voltmeter were used. The calibration constants obtained at 0 dB attenuation were stored in the computer and used to adjust the digital attenuator to obtain the calibrated intensities for each tone.

Throughout the session the electrical signal to the speaker was monitored with an on-line oscilloscope and sounds within the test chamber were monitored with a microphone and amplifier within the chamber connected to a speaker outside the chamber. The opossum's behavior and its pinna alignment with the speaker were also continuously monitored with a television camera mounted just above the speaker and focused on the opossum's head. By these several means, unusual sounds, events, or responses could be immediately detected and analyzed by the experimenter and appropriate actions taken if necessary.

The care and use of the animals reported in this study were approved by the Florida State University's Animal Care and Use Committee which adheres to the guidelines of the Declaration of Helsinki (NIH grant NS7726).

3. Results

The three lowest intensities detected at a statistically reliable rate were used to determine average threshold for each animal at each frequency. The audiograms of five *Monodelphis* opossums, arbitrarily labeled A, B, C, D, and E, are shown in Fig. 1. It can be seen that the audiogram of each opossum generated

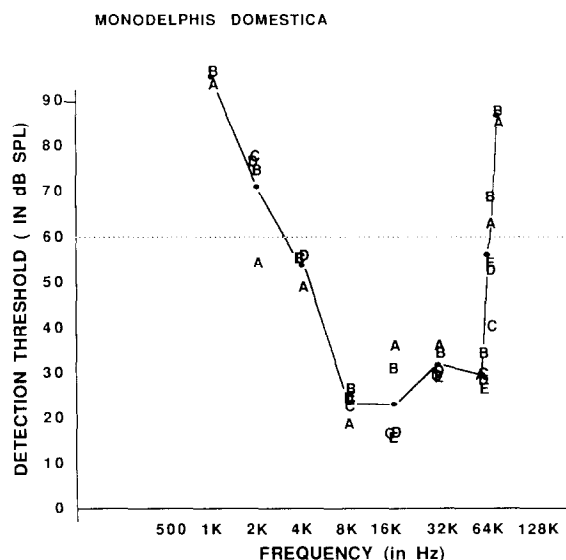


Fig. 1. Audiograms for five *Monodelphis* opossums (letters indicate individual animals). Standard error for each frequency for each animal is smaller than the symbol used. Frequencies above 64 kHz are 75 kHz and 85 kHz. Dashed line indicates 60 dB SPL.

the usual 'U'-shaped curve of detection threshold as a function of tone frequency. Their sensitivity improved as frequency increased above 1 kHz, and the audiogram shows a range of best sensitivity from 8 kHz to 64 kHz. It is notable that these animals were barely able to hear 1 kHz, and then with an average threshold of 95 dB SPL. Thus, it appears that *Monodelphis* is quite insensitive to low frequency tones. Above 64 kHz, the *Monodelphis*' sensitivity decreased rapidly with the two opossums responding to 85 kHz exhibiting an average threshold of 86 dB SPL. At an arbitrary intensity level of 60 dB SPL, a level often used for comparisons among species (Heffner and Masterton, 1980; Heffner and Heffner, 1985b; 1992a), the average hearing range of *Monodelphis* extends from about 3.6 kHz to 77 kHz with an average lowest threshold of 20 dB SPL.

The audiograms of two *Marmosa* opossums labeled A and B are shown in Fig. 2. Once more, the audiograms formed the usual 'U'-shaped function. It can be seen that *Marmosa* is also insensitive to low frequencies – the single subject responding to 1 kHz showing a threshold of 96 dB SPL. Above 2 kHz, sensitivity rapidly improved as frequency increased, and the audiogram shows a range of best sensitivity from 8 kHz to 64 kHz. Above 64 kHz, *Marmosa*'s sensitivity decreased rapidly, with animals responding to 85 kHz at an average intensity of 78 dB SPL. At an intensity level of 60 dB SPL, the average hearing range of *Marmosa* extends from about 3.8 kHz to 80 kHz – both limits not reliably different from those of *Monodelphis*.

Collectively, the results show that neither of these two species is particularly sensitive to tones – the average lowest reliable threshold was 20 dB at 8 and 16

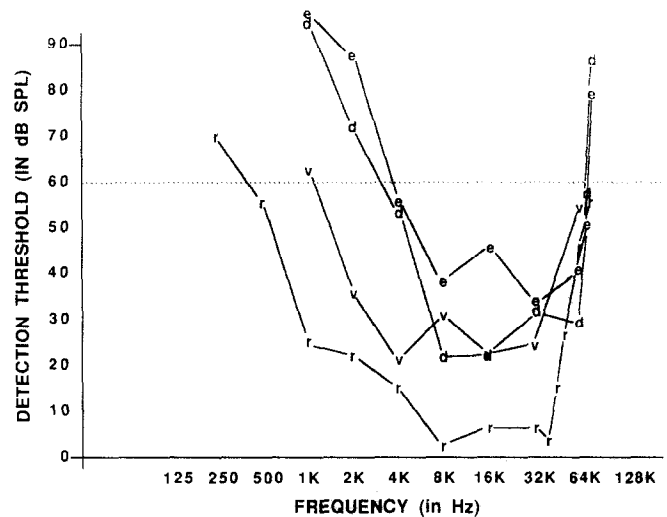


Fig. 3. Averaged audiograms of three species of Didelphid opossums compared to the laboratory rat. d = *Monodelphis domestica*, e = *Marmosa elegans*, v = *Didelphis virginiana* (Ravizza, Heffner and Masterton, 1969), r = rat (Kelly and Masterton, 1977).

kHz for *Monodelphis* and 33 dB at 32 kHz for *Marmosa*. Nevertheless, both species are sensitive to high frequencies (i.e., > 32 kHz), relatively insensitive to low frequencies (i.e., < 8 kHz), and apparently insensitive to frequencies at 1 kHz and below.

In Fig. 3, average audiograms of *Monodelphis* and *Marmosa* opossums are compared to that of the much larger North American opossum (Ravizza et al., 1969) and the laboratory rat (Kelly and Masterton, 1977). The two smaller species of opossums, *Monodelphis* and *Marmosa*, are clearly less sensitive to low frequencies (4 kHz and below) than the larger opossum, *Didelphis virginiana*. However, all three species hear tones of high frequency, although *Monodelphis* and *Marmosa* are slightly more sensitive at 64 kHz than *Didelphis virginiana*.

It can be noted that all three opossums (and particularly *Marmosa*) are relatively insensitive to pure tones throughout their hearing range when compared to a commonplace placental, the laboratory rat. Indeed, the 24 dB SPL average lowest threshold of the three opossums is far higher than the 0.4 dB average lowest threshold of over 40 terrestrial mammals (Heffner and Heffner, 1990a).

4. Discussion

For almost a century the observation that mammals are unique among vertebrates in their possession of a three-ossicle middle ear has been used by paleontologists as a hallmark for a fossil being of mammalian grade. Although the function of this linkage as an impedance-matching device and the implication of

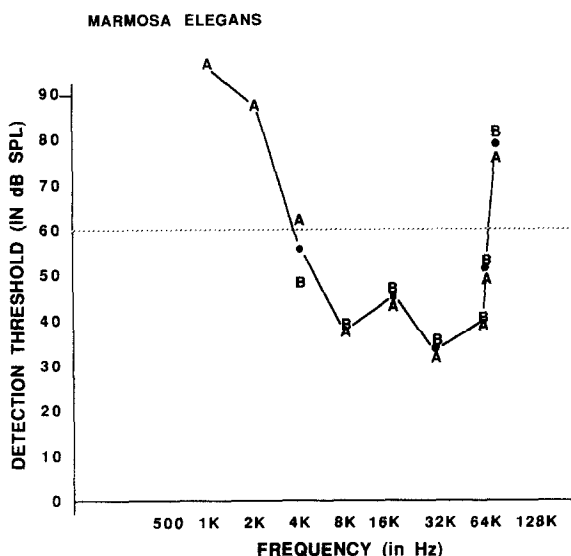


Fig. 2. Audiograms for two *Marmosa* opossums (letters indicate individual animals). Standard error for each frequency for each animal is smaller than the symbol used. Frequencies above 64 kHz are 75 kHz and 85 kHz. Dashed line indicates 60 dB SPL.

impedance-matching for high-frequency hearing has been known for some time, it has become known only recently that high frequency hearing (i.e., above 20 kHz) is the rule among mammals rather than the exception (Masterton et al., 1969; Heffner and Heffner, 1990b). Because the ability of most mammals, and particularly these most primitive of extant mammals, to detect frequencies above 20 kHz is far beyond that achievable by any non-mammalian vertebrate it follows that sensitivity to very high frequencies probably is a very ancient mammalian trait – one following quickly after the acquisition of the 3-ossicle middle ear linkage. It follows that the utility value of high-frequency hearing may have been a primary source of selective pressure for the evolution of the 3-ossicle middle ear to begin with.

It has been argued previously that the selective pressure behind both the morphological transformation and high-frequency hearing may be a reflection of the ecological importance for animals to localize brief sounds, together with the constraint of short interaural distances among small-headed mammals. That is, animals with functionally close-set ears, such as the mouse-sized earliest mammals (Rowe, 1988), may be subjected to a greater selective pressure to hear higher frequencies than animals with more widely set ears because of the inadequacy of interaural time difference cues (Masterton et al., 1969; Heffner and Heffner, 1985a; Heffner and Masterton, 1990).

However, a second line of argument yields another adequate explanation for the same phenomena. It is now commonly accepted that the ancestral middle ear common to marsupials and placentals, and still retained in essential form by opossums, was not surrounded by a rigid wall, but mostly by soft connective tissues (Fleischer, 1973, 1978). According to Fleischer (1978), 'Most troublesome is the fact that the middle ear cavity is not surrounded by a rigid wall, but to a great extent by soft connective tissues. Because of this motions with the head –chewing, swallowing, licking, etc. – will cause slight deformations of the middle ear cavity. This in turn alters the pressure inside and thus changes the sensitivity of the ear. Moreover, this may also alter the stiffness of the ossicular chain with equally undesirable effects. The disturbances mentioned are all at low frequencies, so that a high-frequency ear might not be too affected.' Thus, by this argument, a high-frequency ear is the result of selective pressure for undistorted hearing.

The relative insensitivity of Didelphids when compared to most placentals (see rat, 'R', in Fig. 3, cf. Heffner and Heffner, 1992b) also may be due to the anatomical differences in the structure of the middle ear, this time between opossums and placentals. Unlike the ear of placentals, the tympanic membrane of *Didelphis* is attached at its circumference to a thin tympanic

ring that has no further rigid support of its own (Watson, 1953; Tumarkin, 1955; Fleischer, 1978). This thin tympanic ring in the opossums is one more of the primitive features which have led comparative anatomists and paleontologists to consider the opossum ear as an approximation to a form which was historically intermediate to reptiles and advanced mammals (Le Gros Clark, 1934; Fleischer, 1973, 1978). It is possible that the energy loss at the tympanum may be greater than it is in placentals with their tympanic rings more firmly attached, and this energy loss may account for the lesser sensitivity in Didelphids.

Finally, morphometric measurements of the fossilized middle ear of *Morganucodon*, one of the oldest known mammals (Kermack et al., 1981), suggest that the earliest mammals probably were sensitive to high frequencies though probably not as sensitive to sound as most modern mammals (Rosowski and Graybeal, 1991; Rosowski, 1992). Thus, the opossums in the present sample appear to have retained the primitive characteristics of still more primitive animals by demonstrating sensitivity to high frequencies yet relative insensitivity to sound. Insofar as the small, pouchless South American opossums such as *Monodelphis* and *Marmosa* may be less altered from the primitive form than the North American opossum, *Didelphis virginiana* (Kirsch, 1977; Marshall, 1979; Reig et al., 1987), their still greater bias towards higher frequencies (poorer than *D. virginiana* at 4 kHz and below and slightly better at 64 kHz) may be another reflection of being less advanced in addition to being smaller in size.

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